Out-Back: The Anti-DeSitter Warp

In 1999, Lisa Randall at Princeton University and MIT and Raman Sundrum at Boston University (Figure 23.4) conceived another way to stop gravitational force lines from spreading into the bulk: the bulk could suffer what is called “Anti-deSitter warping.” This warping might be produced by what are called “quantum fluctuations of bulk fields”—but that’s irrelevant to my story so I do not explain it here.[37] Suffice it to say that this mechanism to produce the warping is very natural. By contrast, the Anti-deSitter (AdS) warping itself does not *look* natural at all. It looks downright weird.



Fig. 23.4. Lisa Randall (1962– , right) and Raman Sundrum (1964– , left).

Suppose you’re a microbe, and you live in a face of a microscopic tesseract (Chapter 29). You travel, in your tesseract, out from our brane; perpendicularly out (straight up in Figure 23.5). And suppose you have a microbial pal, who also travels perpendicularly out from our brane. When you and your pal depart our brane, you are 1 kilometer apart (1000 meters; about 0.6 miles). Although you both travel precisely